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Preview of Award 2126334 - Annual Project Report

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Cover

Federal Agency and Organization Element to Which Report is Submitted:	4900
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Project Title:	RCN: Reimagining a Sustainable Data Network to Accelerate Agricultural Research and Discovery
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Accomplishments

*** What are the major goals of the project?**

- Aim 1: Continue developing recommendations, standards, and implementation plans for FAIR data management issues.
  - Aim 1.1: Working group development based on current topics of interest.
  - Aim 1.2: Incentivize working group contributions.
  - Aim 1.3: Organize community workshops and events to enhance AgBioData community development.
- Aim 2: Disseminate project deliverables and recruitment for our network.
  - Aim 2.1: Disseminate AgBioData deliverables and expand increase our network.
  - Aim 2.2: Promote AgBioData and disseminate deliverables via an Ambassador program.
  - Aim 2.3: Provide guidance and recommendations to researchers, journals, editors and reviewers to increase agricultural data FAIRness. Support adoption and application of (AgBioData) FAIR data management standards for scientific literature.
- Aim 3: Disseminate and maintain educational materials to train researchers about FAIR data standards and best practices for Genomics, Genetics and Breeding (GGB) databases.
- Aim 4: Develop the groundwork for sustainable GGB databases.
  - Aim 4.1: Conduct quantitative assessment of member GGB database utilization and impact.
  - Aim 4.2: Identify common sustainability challenges and develop strategic solutions.

- Aim 4.3: Seek new funding for long-term AgBioData database support.

*** What was accomplished under these goals and objectives (you must provide information for at least one of the 4 categories below)?**

Major Activities:

Aim 1.1. Working group (WG) development based on current topics of interest:

We established the Artificial Intelligence Readiness (AIR) WG, in January 2026. This WG has 15 active members and aims to develop best practices for organizing, formatting, annotating, and packaging agricultural data for AI training, validation, and application.

Three established working groups (Fair Scientific Literature (FSL), Single Cell RNAseq Biocuration (SCRNAseq), and NLP for Biocuration) continued working on their objectives (see Significant results).

Aim 1.2. Incentivizing working group contributions:

We awarded three AgBioData Championships (see Significant Results and Impacts).

We also recognized the contributions of community members at our annual conference with awards in the following areas: Mission and Leadership (Leonore Reiser, David Molik, Sunita Kumari), Best Working Group Member (Marcela Tello-Ruiz, Jennifer Clarke, Kapeel Chougule), Fairification (Adam Wright, Laurel Cooper and Kapeel Chougule) and Best Workshop Contributor (Olivia Haley).

Aim 1.3. Organize workshops to enhance AgBioData community development:

We held a hybrid 2026 AgBioData community workshop at Kansas State in Olathe, KS from April 20 to 22, 2026 (see Significant Results).

Aim 2.1. Disseminating AgBioData deliverables and expanding our network:

We participated in five international and national conferences with workshops and hosted booths at PAG and ASPB (see Significant Results and Products).

We published 2 peer-reviewed white papers and submitted a third, which was just accepted (see Products).

We continued expanding our network and engagement with related FAIR Data projects. For example, AgBioData has been invited to become a node of the LIFES initiative (https://www.lifes.institute/).

We recruited new members through conference outreach, webinars, the Ambassador program and social media. We received three membership requests from databases (ManureDB, MARDI and PhiBase). We gained 39 new individual members in the last 12 months. As of July 2026, there were a total of 73 WG participants.

We maintained the website by updating events, adding deliverables and uploading webinar recordings/slides. We sent out quarterly newsletters and maintained an active social media presence on LinkedIn (258 connections, 398 followers). Following guidance from our external advisory committee, we designed changes to our website to improve product visibility, which will be implemented in the coming year.

Aim 2.2. Promoting AgBioData via an Ambassador program

(See Significant Results and Products.)

Aim 2.3. Provide guidance and recommendations to researchers, journals, editors and reviewers to increase agricultural data FAIRness.

The FSL WG extended their recommendations to cover primary repositories (e.g. sequence, proteomics, metabolomics resources). The draft recommendations were released for a 60-day open comment period. The feedback received was incorporated into a table that will accompany the AgBioDatabase Finder tool (https://www.agbiodata.org/finder) on the website.

Aim 3. Disseminate and maintain educational materials to train researchers about FAIR data standards and best practices for GGBs:

The AgBioData Educational Curriculum for FAIR Agricultural Science (https://zenodo.org/records/14278084) has been downloaded over 2,000 times since its publication, with a significant increase in the past year (750 downloads in 2025). We promoted this curriculum at a professional development workshop for Primarily Undergraduate Institutions (PUI) faculty at the ASPB Plant Biology Conference 2026, and solicited proposals for AgBioData Education Implementation Awards. Approximately 50 participants attended the workshop. Our participation generated interest in the curriculum and the newly forming Education and Training Working Group. The Standards for Genetic Variation WG also produced a FAIR User Guide for the Adoption of Standard SNP Identifiers (https://www.agbiodata.org/FAIR_user_guide_rsID_adoption), extending the group's technical recommendations into an adoption and teaching resource.

Aim 4.1. Conduct quantitative assessment of member GGB database utilization and impact.

We completed a pilot project to identify and collect key metrics to quantify the impact of AgBioData databases, including economic value of database species, usage, citations (primary and secondary), and calculation of return on investment (ROI) on database funding. To identify the best approach for scaling this assessment, the team benchmarked several AI and text-mining tools for automated citation and impact-metric collection (including Google Scholar, Europe PMC, and Publish or Perish), weighing completeness of results against practical usability. This evaluation informed the development of TRACE (Tracking Research And Citation Evidence), a Python-based citation analysis pipeline (built with Claude AI, run via Google Colab, requiring no local installation or coding expertise) that searches OpenAlex, Semantic Scholar, Europe PMC, and PubMed using complementary URL-mention and citation-graph methods, deduplicates results, filters out database-team publications, and exports APA-formatted citation lists. Initially developed for GDR and adapted for MaizeGDB, the pipeline is designed to be reconfigured for any member database and is currently in alpha evaluation. Applied to Gramene, it identified 523 citing papers (2020–2024) across 47 journals, yielding a database-specific ROI estimate of >40,000:1 (consortium-wide estimate: >24,000:1).

Aim 4.2. Identify common sustainability challenges affecting GGBs and make recommendations for creative and strategic solutions.

As a result of extensive networking, we have also been asked to participate in an ARS approved meeting between the NRSP 8 Animal Genetics and the NRSP9 Animal Nutrition programs, a first of its kind in our field.

Aim 4.3. Seek new funding for long-term AgBioData database support:

To secure the future of AgBioData and improve the sustainability of member databases, we established a Grants Subcommittee that has drafted a strategic plan that aligns

consortium goals, research aims and funding opportunities. The plan considers a diversified portfolio and potential organizational and governance shifts.

Specific Objectives: The specific objectives we outlined for year 5 include

- Aim 1.1: We continued three ongoing issue-focused working groups and established one new working group based on community feedback (i.e. AI readiness working group).
- Aim 1.2: We continued incentivizing participation, promoting leadership, and supporting professional development via the Champion Programs.
- Aim 1.3: We organized a hybrid Annual Community Workshop (K-State Olathe campus, April 20-22, 2026) to bring together the AgBioData community to share ideas and build community. We continued to host monthly webinars.
- Aim 2.1: We continued actions to disseminate our results, and expand our network and participants. Such actions included participation in international and national conferences, reaching out to relevant databases and key stakeholders in the database and FAIR data ecosystem, and expanding our network.
- Aim 2.2: We continued to select and support AgBioData Ambassadors to promote AgBioData and disseminate results.
- Aim 2.3.1: We provided guidance and recommendations to researchers, journals, editors and reviewers to increase agricultural data FAIRness. Specifically, we drafted guidance for data type specific repositories.
- Aim 2.3.2: We maintained and updated the AgBioData website to make deliverables more accessible.
- Aim 3: We disseminated the AgBioData curriculum and supported its expansion by offering Education Awards.
- Aim 4: We continued developing the groundwork for sustainable Genetic, Genomic and Breeding Databases via impact assessments and strategic reports targeted to various stakeholders. We hosted a Sustainability workshop at the PAG33 conference aimed at devising long- and short-term strategies, and started a working group to research and pursue consortium funding.

Significant Results: **Aim 1.1: Issue-focused working groups:**

The NLP4Biocuration WG drafted a white paper summarizing their assessments of various NLP-based tools for biocuration. Using a donated server from Phoenix Bioinformatics, Dr. Muskan Kapoor, and Champion Dr. Laura Cotes Tibbs (see Impacts) deployed a test instance of Texpresso. The AgBioData Texpresso has been populated with papers from five member databases, and its API has been updated to enable improved integration with model organism databases and their existing curation workflows. MaizeGDB is currently evaluating a maize-specific prototype.

The SGV WG published a manuscript (see Products) describing a survey of current standards for managing agricultural genetic variation and recommending the adoption of reference SNP identifiers (rsIDs) as a key step to promote FAIR data practices. Milestones reported included introducing ~220 million rsIDs to Gramene and pangenome databases, projecting rsIDs from reference to pangenome varieties in sorghum and maize, and developing an agricultural FAIR user guide for rsID adoption. Dr. Tello-Ruiz, presented the achievements and recommendations of the WG at PAG33 (see Products).

The FSL WG compiled a table of recommended primary repositories to complement our finder tool. We solicited community feedback during a 60-day open comment period. The table is available on our website (https://www.agbiodata.org/db-finder-repositories).

The Single-Cell RNAseq WG proposed an implementation framework centered on four priorities: harmonized metadata standards, evidence-based cell-type annotation, benchmark datasets with reproducible workflows, and community engagement. This framework will form part of a community roadmap paper outlining a federated ecosystem that connects existing resources through shared standards, interoperable infrastructure, and sustained collaboration across agricultural systems.

Aim 1.2: Incentivize Working Group participation

Three Champion awards were made, two to early career researchers (see Impacts). Dr. Tanya Berardini is co-Chair of the NLP for Biocuration working group, which has successfully achieved its deliverables, including an analysis of NLP-assisted biocuration tools, a test deployment of Texpresso, and a white paper. Dr. Laura-Tibbs Cortes (postdoctoral researcher, NLP for Biocuration WG) built and populated the Texpresso instance and is testing it at MaizeGDB. Dr. Olivia Haley (postdoctoral researcher) is a co-Chair of the newly formed AI readiness working group. Dr. Haley also became the first postdoctoral scholar to serve on the Steering Committee (see Impacts). We met quarterly with Champions to assess their progress and support them through challenges. Champions are required to write a blog or social post about their contributions (e.g., https://www.agbiodata.org/node/647).

Aim 1.3: Community Workshop

We held a hybrid Community Workshop at Kansas State-Olathe from April 20 to 22, 2026, that included WG presentations, a panel discussion, an international-focused session, awards, and breakout room discussions on topics and questions crucial to the WGs. Slides (https://www.agbiodata.org/2026-agbcomworks) and recordings (https://tinyurl.com/4yvmw86b) are available. A total of 76 individuals registered, with 17 attending in person and at least 25 participating virtually each session. Post-workshop evaluations showed high satisfaction, with over 96% of 26 respondents reporting they were very satisfied or satisfied overall. The workshop also strengthened community engagement and awareness: 85% of respondents indicated they would attend future AgBioData meetings, 81% reported an improved understanding of the AgBioData working groups, 69% gained greater clarity on the relevance of AgBioData to their own work, 65% would encourage others in their network to join, and 45% reported expanding their professional network through the workshop. A summary of the workshop is being prepared for submission to GenRes Journal.

Aim 2: Dissemination of results, broadening the network

Aim 2.1: Following extensive delays due to new USDA guidelines on co-authorship, a manuscript summarizing the accomplishments, lessons learned, and key recommendations from the first 3 years of the RCN grant (“The Future is FAIR: A Community Framework for Enhanced Data Management and Data Sharing”) has been accepted by the journal Database (see Products). We hosted 11 webinars averaging 17 participants. Recordings were posted to our website and YouTube. Subscribers to our YouTube channel increased to 211, with an average of 31 views per webinar.

To disseminate our results and increase our network, we organized three external workshops. At the Plant & Animal Genome conference (PAG 33; January 9-14, 2026), we presented progress and recommendations from our working groups at the

"AgBioData Consortium: Challenges and Recommendations for FAIR Genetic, Genomic and Breeding Data". The AgBioData Database Sustainability Workshop brought together database users, managers, and developers in an interactive world-café format. Participants engaged in collaborative discussions and brainstorming sessions to identify and explore viable solutions for the long-term database sustainability. We hosted a booth and engaged with nearly 200 individuals. At the ASPB Plant Biology Conference (July 19, 2026), we organized a workshop "Plant Bioinformatics Resources for FAIR Agricultural Data Discovery and Reuse " that featured an introduction to AgBioData and overview presentations from member digital resources. The workshop was attended by ca. 60 people. We also hosted a booth with representatives from member databases, who attended and interacted with community members.

Aim 2.2: 7 Ambassadors presented talks or posters at four conferences (see Products). Sushma Naithani attended the 19th Annual International Biocuration Conference (Cape Town, Africa, April 20–24, 2026). She presented a talk and participated in a panel discussion that included an overview of the AgBioData Consortium, the adoption of FAIR data principles, opportunities for building collaborations with other biological databases, and strategies for training and outreach for students and early-career scientists. Kapeel Chougule attended Advances in Genome Biology and Technology in Agriculture (AGBT-Ag, Phoenix, Arizona, April 12-15, 2026). He presented a poster and flash talk on "NLP4Biocuration: Advancing FAIR, AI-Assisted Literature Curation Across the AgBioData Consortium". Sunita Kumari co-organized the AgBioData workshop at the ASPB2026 conference (Ottawa, CA; July 18-22, 2026) and represented AgBioData at the booth. Four ambassadors presented talks in the AgBioData workshop at PAG33 (San Diego, CA; January 9-12, 2026): John McNamara (SWG), David Molik (Organizer and FSLWG), Muskan Kapoor (scRNAseq WG), and Tanya Berardini (NLP for Biocuration).

Aim 2.3 See 'Major activities'.

Aim 3: Disseminating and maintaining educational materials

See 'Major activities'.

Aim 4: Sustainability

Aim 4. 1. Conduct quantitative assessment of member GGB database utilization and impact. See 'Major activities'.

Aim 4.2. Identify common sustainability challenges affecting GGBs and make recommendations for creative and strategic solutions.

The Sustainability Working (SWG) drafted a white paper and 4 advocacy statements detailing the national importance of agricultural databases, and presented them to the AgBioData Community for feedback.

Aim 4.3. Seek new funding for long-term AgBioData database support. See 'Major activities'.

Key outcomes or Other achievements:

Other Achievements

Evaluation by the External Advisory Committee

The Steering Committee (SC) met with the External Advisory Committee (EAC) in October 2025 to present progress and challenges during year 4 and obtain feedback and suggestions for improvement. The EAC consists of internationally recognized experts in data management. The current members are Anne Kwitek (Medical College of Wisconsin), Lindsey Anderson (Pacific Northwest National Lab), James Koltes (Iowa State University), Robert Cottingham (Oak Ridge National Lab), and Susanna Sansonne (University of Oxford). The EAC's strategic guidance for AgBioData leadership and AgBioData Working Group included ways to 1) improve data FAIRness and connectivity among genomic, genetic, and breeding (GGB) databases, 2) enhance community engagement for recruiting new members and collaborative partners, and 3) sustain AgBioData financially through specific grant programs. Specific recommendations included seeking ways to standardize and consolidate some aspects of member databases for cost effectiveness and long-term sustainability.

Program coordination

The new PC, Dr. Marcela K. Tello-Ruiz, coordinated the AgBioData WGs and engaged in outreach and educational activities. She was the leading and corresponding author of the white paper urging the agricultural community to adopt standard rsIDs as SNP identifiers. Together with a workshop coordination sub-committee, she organized three workshops, including the annual AgBioData community workshop, the AgBioData workshop at the PAG33 conference, and the Plant Bioinformatics Resources workshop at the 2026 ASPB conference. She manages the social media accounts and website, and coordinates the AgBioData monthly webinars by inviting and providing support to the speakers. She also provides extensive support to the steering committee.

The following changes were made to the Steering Committee:

Emily Clark (EMBL-EBI), Genome Analysis Team Leader, and representative of Ensembl, an international AgBioData member database (replaces Sarah Dyer); Olivia Haley (USDA ARS), ORISE Postdoctoral Research Fellow, MaizeGDB contributor. She co-chairs the AI-Readiness WG, and has been key in launching the Education and Training WG. Peter Selby and John McNamara also completed their respective terms as Steering Committee members.

*** What opportunities for training and professional development has the project provided?**

Training and professional development opportunities have primarily been provided to the project participants (Aim 1.1, 1.2, 2 and 3) and the PC. Participants in the working groups include early-career researchers who are developing leadership skills and are co-authors on the white papers and products generated during this RCN.

Olivia Haley (USDA ARS) is a USDA ORISE postdoctoral fellow. She played a significant role in organizing the hybrid conference, helping to set the agenda, run the workshop and conduct evaluations. She is now the first postdoctoral scholar to become a member of the AgBioData steering committee and participates in the future funding/grantwriting team where she is developing her grantwriting skills. She also co-chairs the AIR WG with Sunita Kumari. There she has led the group's early efforts to define priorities, establish milestones, and begin refining AI-readiness guidance for agricultural biological databases. She coordinates regular meetings, maintains accountability for action items, and is guiding the development of an AI-readiness assessment framework and tool to support AI adoption across the AgBioData community.

Laura Tibbs-Cortes (USDA ARS) is a USDA ORISE postdoctoral scholar with MaizeGDB and a member of the NLP for Biocuration working group. She is a Champion of that working group. As champion, she has built a shared test instance of Textpresso for AgBioData and developed a maize-specific prototype to support machine-assisted literature curation. She has coordinated the integration of ontologies, assembled curated maize literature for testing, and is working with MaizeGDB to evaluate and refine the platform for broader community use.

David Molik (Kansas State University), an Early Career Researcher, is an Associate Professor and Librarian, Lead Library Research Officer, and Head of the Open Publishing Exchange, which includes New Prairie Press, at Kansas State University. He was elected to the AgBioData Steering Committee in April 2025 and also serves on the FAIR Scientific Literature Working Group. Dr. Molik organized and hosted the 2026 hybrid AgBioData community meeting after it was relocated to Kansas State University. As part of that transition, the project subcontract moved from Washington State University to K-State, and Dr. Molik joined the grant as a co-PI and serves as PI of the K-State subcontract. In this role, he manages the K-State budget, coordinates the local team, contributes to meeting and project planning, and helps establish goals and monitor progress. He also secured external support for the community meeting, reducing costs to the project. As a co-PI, Dr. Molik participates in project-wide budget discussions, planning, priority setting, and assessment of progress toward AgBioData's goals.

In addition to the support provided by the PC and the members of the steering committee when needed, working group members who are selected as AgBioData Champions receive additional support. For many of the WG participants who are not formally trained in data management (i.e., data-generating scientists), they also receive training as they learn from their WG colleagues. Finally, all working group participants receive recognition for their work on the AgBioData website (https://www.agbiodata.org/current-working-groups) and several individuals receive additional honors (via community awards) that can be included in their CVs. Service to AgBioData can also be included in their ORCID profiles.

The PC, Dr. Marcela K. Tello-Ruiz (Phoenix Bioinformatics), has a background in genetics, comparative genomics and bioinformatics, and received training and professional development in project management and community engagement, through a Project Management for Scientists Bootcamp from the Center for Scientific Collaboration and Community Engagement. Before starting as the PC, Marcela completed the activities as co-Chair of the Standards for Genetics Variation WG and was first and a corresponding author on the working group paper. This was the first manuscript she led from beginning to end. She also received the 2026 AgBioData Mission and Leadership, as well as the Best Working Group Member awards.

*** Have the results been disseminated to communities of interest? If so, please provide details.**

We have disseminated our results through a variety of modes including the website, newsletters, social media, publications and events.

Our website (https://www.agbiodata.org/) is actively maintained to disseminate our activities and products. Dr. Tello-Ruiz has updated content throughout to keep the site current with events, including upcoming webinars and conferences; educational materials such as new publications and user guides; webinar recordings on YouTube; downloadable slide presentations; etc. We use our mailing lists, quarterly newsletters, social media platforms (primarily LinkedIn), and Slack channels to disseminate news, results, and products. We continue to host monthly webinars on various topics, including data management and biocuration, and community building.

In Year 5 we published 2 peer-reviewed white papers; a third was submitted and was just accepted.

- Tello-Ruiz, M.K.. et al. Adoption of Standard Reference SNP Identifiers in Agricultural Genomics for Interoperability and Data Reuse. Sci Data 13, 885 (2026). https://doi.org/10.1038/s41597-026-07208-0
- Callwood, J., et.al, The need for robust, FAIR phenomic databases supporting agricultural efficiency and resiliency, Science and Public Policy, 52 (2025) https://doi.org/10.1093/scipol/scaf039
- Marrano, A. et al. The Future is FAIR: A Community Framework for Enhanced Data Management and Data Sharing. Submitted to Database (Accepted) .

As of August 2026, our 12 published works have been cited a total of 213 times in 181 papers (see attached PDF agbiodatareport.pdf).

In Year 5, we organized three external workshops:

- The in-person workshop “The AgBioData Consortium: Challenges and Recommendations for FAIR Genetic, Genomic and Breeding Data” at the Plant & Animal Genome conference (PAG 33; January 9-14, 2026; ~40 participants) addressed(1) challenges associated with sustainability of agricultural databases and presented recommendations, and resources developed by the working groups and (2) provide preliminary insights on the long-term sustainability of the AgBioData

database resources. Abstracts and slides of the working groups' presentations are available at https://www.agbiodata.org/agb-pag33.

- The in-person AgBioData Database Sustainability Workshop at PAG 33 brought together ~20 database users, managers, and developers in an interactive world-café format. Participants engaged in collaborative discussions and brainstorming sessions to identify and explore viable solutions for the long-term sustainability of AgBioData's GGB databases.
- The in-person workshop "Plant Bioinformatics Resources for FAIR Agricultural Data Discovery and Reuse " at the ASPB Plant Biology Conference (July 19, 2026) introduced and updated approximately 60 plant biologists on a variety of digital bioinformatics resources and featured overview presentations from the AgBioData member digital resources.

In addition to the AgBioData-hosted workshops, we represented AgBioData at the following events:

- The 19th Annual International Biocuration Conference, held in Cape Town from April 20–24, 2026: Sushma Naithani, a Steering Committee member of the AgBioData Consortium, presented a talk and participated in a panel discussion that included an overview of the AgBioData Consortium, the adoption of FAIR data principles, opportunities for building collaborations with other biological databases, and strategies for training and outreach for students and early-career scientists.
- Advances in Genome Biology and Technology in Agriculture (AGBT-Ag;) - Muskan Kapoor presented a Poster and flash talk on the NLP for Biocuration working group. "NLP4Biocuration: Advancing FAIR, AI-Assisted Literature Curation Across the AgBioData Consortium".
- ASPB Primarily Undergraduate Institution (PUI) Faculty Professional Development Workshop, Harnessing Artificial Intelligence (AI) to Advance Research Productivity and Integrate Research-Based Teaching at PUIs. July 18, 2026. Dr. Tello-Ruiz gave a brief presentation about the AgBioData Curriculum and participated in a panel discussion on AI and Plant Biology Education.

*** What do you plan to do during the next reporting period to accomplish the goals?**

Aim 1: Continue developing recommendations, standards, and implementation plans for FAIR data management issues.

Aim 1.1: The following issue specific working groups will continue and should complete their objectives in the next year: The Single Cell Biocuration will complete their white paper. The Natural Language Processing for Biocuration will complete their white paper and evaluation of the test instance of Texpresso. The AI-Readiness WG will complete their AI readiness self-assessment checklist and their evaluation of Model Context Protocol (MCP) servers at AgBioData databases.

Aim 1.2: To promote and reward leadership within the WGs, we will continue to support our "AgBioData Champions" aiming to encourage WG participation, the development and accomplishment of significant WG deliverables (e.g., white papers), and outreach initiatives targeting key stakeholders. We will continue quarterly meetings with the AgBioData Champions, track their progress, and provide support and feedback as needed.

Aim 1.3: We plan to hold a virtual AgBioData community workshop in 2027. Holding annual AgBioData meetings, where consortium members convene and share updates, has been essential for maintaining the AgBioData community's motivation, re-energizing WG members and cross fertilization of ideas across WGs. The AgBioData community workshops will provide a forum for brainstorming new, emerging data-related issues (see Aim 1.1) and sharing the developed guidelines. Along with progress reports from the WGs, breakout sessions on questions from the WGs, we plan to invite guest speakers to seed discussions.

Aim 2: Disseminate project deliverables and recruitment for our network.

We will incorporate actions to further expand the AgBioData community, which comprises agricultural database creators, users, and managers. We will continue to coordinate with existing organizations (e.g., AG2PI, Plant Cell Atlas) and professional societies (e.g., ASPB, SACNAS) to recruit key stakeholders for the community and define collaborative opportunities. We will also reach out to database resources not part of the consortium yet and provide support to the active WGs in need of new members. We will continue to support Ambassadors who will promote AgBioData and disseminate our deliverables. We will host an AgBioData workshop at PAG34 in January 2027 and support participation of at least one

Ambassador per working group to present. We will also complete our website redesign plan to put our deliverables such as the FAIR Ag Curriculum, front and center. We will continue to recruit and support AgBioData Ambassadors to disseminate our results broadly.

The FSL WG will (1) finalize documenting recommendations for core metadata/data for key datatypes hosted by AgBioData GGBs and (2), identify scholarly publishers/society journals who will partner to promote these standards as part of their review and publishing process. This supports emerging federal requirements for data submission at the time of publication and AI readiness by ensuring open data is also FAIR.

Aim 3: Disseminate and maintain educational materials to train researchers about FAIR data standards and best practices for GGB databases.

This aim will be a key focus of the coming year. In the Fall of 2026, we will launch a new Education and Training WG to develop, improve, and expand educational resources for researchers, students, and professionals working with agricultural biological databases. The group will also coordinate with partner institutions and AgBioData member databases to design, deliver, and evaluate accessible online learning experiences tailored to the needs of agricultural biological database users. We have expanded the scope of the previous Education WG with new members and new leadership. The goals of the new WG are to (1) critically review and update the current modules to better serve target audiences, including students and postdocs, and senior scientists, (2) survey the needs of the community (PUIs, land grants, etc.), and (3) develop new training materials/modules by collaborating with the member databases to support undergraduate curriculum and training of the agricultural genomics community.

Aim 4 Sustainability: We will finalize the current TRACE tool prototype and make it available to AgBioData member databases; and host a database sustainability workshop at the Plant and Animal Genome Conference in January 2027. We will assess current goals for the sustainability working group, and strategically develop the new direction for this working group.

Supporting Files

Filename	Description	Uploaded By	Uploaded On
agbiodatareport.pdf	Screenshot of the report generated by our citation/impact tool using AgBioData papers as the seed.	Leonore Reiser	08/26/2026

Products

Books

Book Chapters

Inventions

Journals or Juried Conference Papers
View all journal articles and conference papers currently available in the NSF Public Access Repository for this award.

The results in the NSF Public Access Repository will include a comprehensive listing of all journal articles and conference papers recorded to date that are associated with this award.

Hafner, Alenka and DeLeo, Victoria and Deng, Cecilia H and Elsik, Christine G and S Fleming, Damarius and Harrison, Peter W and Kalbfleisch, Theodore S and Petry, Bruna and Pucker, Boas and Quezada-Rodríguez, Elsa H and Tuggle, Christopher K and Koltes, James E. (2025). Data reuse in agricultural genomics research: challenges and recommendations. *GigaScience*. 14 . Status = Added in NSF-PAR PAR ID: 10585652

Federal Government's License = Acknowledged. (Completed by Reiser, Leonore on 04/28/2025) Full text Citation details

Marrano, Annarita and Cabugos, Leyla and Hafner, Alenka and Kapoor, Beant and McNamara, John and O'Donnell, Megan and Reiser, Leonore and Tello-Ruiz, Marcela Karey and Zhang, Huiting and Staton, Margaret. (2025). A teaching and training framework to promote findable, accessible, interoperable, and reusable data generation in agriculture. *Database*. 2025 . Status = Added in NSF-PAR PAR ID: 10585025

Federal Government's License = Acknowledged. (Completed by Reiser, Leonore on 04/25/2025) Full text Citation details

Cannon, Ethalinda KS and Molik, David C and Wright, Adam J and Zhang, Huiting and Honaas, Loren and Chougule, Kapeel and Dyer, Sarah. (2025). Guidelines for gene and genome assembly nomenclature. *GENETICS*. 229 (3) . Status = Added in NSF-PAR PAR ID: 10584718

Federal Government's License = Acknowledged. (Completed by Reiser, Leonore on 04/24/2025) Full text Citation details

Deng, Cecilia H and Naithani, Sushma and Kumari, Sunita and Cobo-Simón, Irene and Quezada-Rodríguez, Elsa H and Skrabisova, Maria and Gladman, Nick and Correll, Melanie J and Sikiru, Akeem Babatunde and Afuwape, Olusola O and Marrano, Annarita and Rebollo, Ines and Zhang, Wentao and Jung, Sook. (2023). Genotype and phenotype data standardization, utilization and integration in the big data era for agricultural sciences. *Database*. 2023 . Status = Added in NSF-PAR PAR ID: 10486308

Federal Government's License = Acknowledged. (Completed by Reiser, Leonore on 01/17/2024) Full text Citation details

Clarke, Jennifer L and Cooper, Laurel D and Poelchau, Monica F and Berardini, Tanya Z and Elser, Justin and Farmer, Andrew D and Ficklin, Stephen and Kumari, Sunita and Laporte, Marie-Angélique and Nelson, Rex T and Sadohara, Rie and Selby, Peter and Thessen, Anne E and Whitehead, Brandon and Sen, Taner Z. (2023). Data sharing and ontology use among agricultural genetics, genomics, and breeding databases and resources of the Agbiodata Consortium. *Database*. 2023 . Status = Added in NSF-PAR PAR ID: 10486307

Federal Government's License = Acknowledged. (Completed by Reiser, Leonore on 01/17/2024) Full text Citation details

Callwood, Jodi and Celebioglu, Burcu and Gladman, Nicholas and Jung, Jinha and Lachowiec, Jennifer and Quezada_Rodriguez, Elsa H and McNamara, John P and Clarke, Jennifer. (2025). The need for robust, FAIR phenomic databases supporting agricultural efficiency and resiliency. *Science and Public Policy*. . Status = Added in NSF-PAR PAR ID: 10630955

Federal Government's License = Acknowledged. (Completed by Reiser, Leonore on 08/27/2025) Full text Citation details

Tello-Ruiz, Marcela K and Cezard, Timothee and Andorf, Carson and Balyan, Sonia and Bassil, Nahla V and Beier, Sebastian and Bushakra, Jill M and Chang, Tao-Ho and Chogule, Kapeel and Cobo-Simón, Irene and Dyer, Sarah and Elsik, Christine G and Gladman, Nicholas and Harrison, Melanie and Humann, Jodi and Kim, Catherine and Kumar, Vivek and Nandety, Raja S and Nelson, Rex and Olson, Andrew and Ware, Doreen. (2026). Adoption of Standard Reference SNP Identifiers in Agricultural Genomics for Interoperability and Data Reuse. *Scientific Data*. . Status = Added in NSF-PAR PAR ID: 10699980

Federal Government's License = Acknowledged. (Completed by Reiser, Leonore on 07/23/2026) Full text Citation details

Cabugos, Leyla and Buble, Katheryn and Daenzer, Jenna and Jung, Sook and Main, Dorrie and Marrano, Annarita and Molik, David and Raciti, Daniela and Wright, Adam and Yook, Karen and Reiser, Leonore. (2025). AgBioDatabase Finder: an online tool to help researchers find and submit agricultural genomic, genetic, and breeding data. *microPublication biology*. 2025 . Status = Added in NSF-PAR PAR ID: 10677543 Type: Version of Record

Federal Government's License = Acknowledged. (Completed by Reiser, Leonore on 04/16/2026) Full text Citation details

The need for robust, FAIR phenomic databases supporting agricultural efficiency and resiliency. Science and Public Policy (2025) Jodi Callwood, Iowa State University Burcu Celebioglu, Oregon State University Nicholas Gladman, USDA Agricultural Research Service Jinha Jung, Purdue University Jennifer Lachowiec, Montana State University Elsa Quezada Rodriguez, Universidad Nacional Autonoma de Mexico John McNamara, Washington State University DOI:10.1093/scipol/scaf039. Status = PUBLISHED.

The Future is FAIR: A Community Framework for Enhanced Data Management and Data Sharing. Maranno, A. et al., (Under review at Databases) Abstract The AgBioData consortium is a community-driven initiative supporting over 40 genomic, genetic, and breeding (GGB) databases for agricultural research. Informed by the FAIR (Findable, Accessible, Interoperable,

and Reusable) data principles, AgBioData aims to improve data management, sharing, and reuse across diverse biological data types. Through a National Science Foundation Research Coordination Network grant (award #2126334), the consortium engaged stakeholders via working groups, webinars, and community workshops to identify key data challenges and develop recommendations, tools, and training materials. Major outcomes include guidelines for standardizing genetic and genomic data, SNP harmonization protocols, recommendations for ontology usage, and an open-access curriculum for FAIR data literacy. Other key topics addressed in the first three years include data federation, improved single-cell data annotation, and advanced database interoperability. Surveys indicate an increase in community awareness and adoption of FAIR practices. AgBioData represents a scalable model for enhancing agricultural data infrastructure and sustainability, with ongoing efforts focused on AI-readiness and the long-term support of curated data resources.. Status = ACCEPTED.

Licenses

Other Conference Presentations / Papers

Leonore Reiser, Leyla Cabugos, Katheryn Buble, Jenna Daenzer, Sook Jung, Dorrie Main, David Molik, Daniela Raciti, Adam Wright, Karen Yook and Annarita Marrano (2026). *AgBioDatabase Finder: An Online Tool to Help Researchers Find and Submit Agricultural Genomic, Genetic, and Breeding Data*. International Plant and Animal Genome Conference (PAG33). San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Sarah Dyer, Marcela K. Tello-Ruiz, Sunita Kumari, Leonore Reiser and AgBioData Consortium (2026). *Building AI-Ready Agricultural Data: A New Initiative from the AgBioData Consortium*. International Plant and Animal Genome Conference (PAG33). San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Megan O'Donnell and Leyla Cabugos (2023). *Building a Community Lead FAIR Data Curriculum for Agricultural Biology Data*. Research Data Access and Preservation (RDAP) Summit 2023. Virtual. Status = OTHER; Acknowledgement of Federal Support = Yes

Annarita Marrano,Darwin Campbell,Jaqueline Campbell,Ethalinda Cannon,Laurel Cooper,Peter Harrison,Lisa Harper,Eva Huala,Sook Jung,Sunita Kumari,John Mcnamara,Sushma Naithani,Monica Poelchau,Leonore Reiser,Margaret Staton,Marcela K Tello-Ruiz (2023). *Can we make phenomic, genetic, and genomic data FAIR? The AgBioData road trip towards best practices of data sharing and management in agricultural research and education*. The North American Plant Phenotyping Network (NAPPN) Annual Conference 2023. St. Louis, MO. Status = OTHER; Acknowledgement of Federal Support = Yes

Sook Jung, Jodi Humann, Jing Yu, Ping Zheng, Taein Lee, Chun-Huai Cheng, Katheryn Buble, Heidi Hough, Dorrie Main (2022). *Complementary Efforts By Other Databases, Initiatives and Networks to Facilitate Use of the Existing Genetic Diversity in the NPGS System and Other Collections (GDV, GDR, AgBioData)*. American Society for Horticultural Science 2022 Annual Conference. Chicago, IL. Status = OTHER; Acknowledgement of Federal Support = Yes

John McNamara, Marcela K. Tello-Ruiz and Dorrie Main (co-organizers) (2026). *Data Resource Sustainability and Funding Workshop*. International Plant and Animal Genome Conference (PAG33). San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Alenka Hafner, James E. Koltes (2024). *Data Reuse in Agricultural Genomics Research: Challenges, Solutions, and a Case Study*. Plant & Animal Genome Conference/PAG31. San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Muskan Kapoor, Jayne Wiarda, Crystal L. Loving, Sunita Kumari, Doreen Ware and Christopher K. Tuggle (2026). *Developing FAIR Open-Source Interactive Web Browsers for Individual Labs to Explore Single-Cell Data: Shiny-Piggi As an Example from the Agbiodata Single Cell Working Group*. International Plant and Animal Genome Conference (PAG33). San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

David Molik, Katheryn Buble, Leyla Cabugos, Jaqueline Campbell, Jenna Daenzer, Daniela Raciti, Leonore Reiser, Sook Jung, Adam Wright and Karen Yook (2024). *FAIR Scientific Literature Is Not What You Think: How to Know Where Data Goes*. Plant & Animal Genome Conference/PAG31. San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Alberto Cámara (2024). *FAIRification of Germplasm Bank Resources*. Plant & Animal Genome Conference/PAG31. San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Marcela K. Tello-Ruiz, Timothee Cezard, Sebastian Beier, Rex Nelson, Jill Bushakra, Jodi L. Humann, Sarah Dyer, Osman Ariel Gutierrez, Nahla Bassil, Sonia Balyan, Sharon Wei and Doreen Ware (2026). *From Fragmented Data to Shared Knowledge: Adopting rsIDs in Agricultural Genomics*. International Plant and Animal Genome Conference (PAG33). San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Peter C. Selby (2024). *How to Implement Practical Data Federation: Technology Review and Training Material*. Plant & Animal Genome Conference/PAG31. San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

John McNamara (2026). *Integrating Phenomics and Phenotypes into Agricultural Databases*. International Plant and Animal Genome Conference (PAG33). San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Joshua W Young (2024). *Mapping Possible Paths Towards Financial Sustainability for Digital Research Resources*. Plant & Animal Genome Conference/PAG31. San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Tanya Z. Berardini (2026). *NLP4Biocuration Working Group Update: Shared Learning and Knowledge Sharing*. International Plant and Animal Genome Conference (PAG33). San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Tanya Z. Berardini (2026). *NLP4Biocuration Working Group Update: Shared Learning and Knowledge Sharing*. International Plant and Animal Genome Conference (PAG33). San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Kapeel Chougule, Adam Wright, Srikanth Kumar Karaikal, Alyssa Proia, Tanya Z. Berardini, Sushma Naithani, Parul Gupta, Sook Jung, Elly Poretsky, Laurel Cooper, Edwin Jun Kiat Ong,Taner Z. Sen, Laura E. Tibbs-Cortes, Olivia C. Haley, Carson M. Andorf, Rex T. Nelson, Lydia G. Tressel, Qi Li, Doreen Ware, Jacqueline D. Campbell, Marcela K. Tello-Ruiz (2026). *NLP4Biocuration: Advancing FAIR, AI-Assisted Literature Curation Across the AgBioData Consortium*. Advances in Genome Biology and Technology (AGBT) Agriculture. Phoenix, AZ. Status = OTHER; Acknowledgement of Federal Support = Yes

Marcela K. Tello-Ruiz and Sunita Kumari (co-organizers) (2026). *Plant Bioinformatics Resources for FAIR Agricultural Data Discovery and Reuse Workshop*. Plant Biology 2026. Ottawa, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Dorrie Main (2024). *Research Impact of Plant Community Databases*. Plant Biology 2024. Honolulu, HI. Status = OTHER; Acknowledgement of Federal Support = Yes

Marcela Karey Tello-Ruiz (2024). *Standardizing Biocuration of Genetic Variation Data to Promote FAIRification*. Plant & Animal Genome Conference/PAG31. San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Marcela K. Tello-Ruiz (2026). *Stretching Genomics Budgets with rsIDs: Practical Lessons from Agronomic Crops for Efficient SNP Calling In Niche Species*. International Plant and Animal Genome Conference (PAG33). San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Marcela K. Tello-Ruiz (2025). *The AgBioData Consortium: Enabling FAIR Agricultural Data through Standardized Protocols and Practices*. VI International Congress of the Peruvian Society of Bioinformatics and Computational Biology. virtual. Status = OTHER; Acknowledgement of Federal Support = Yes

Jaqueline Campbell, Peter W. Harrison, Sook Jung, Sunita Kumari, Dorrie Main, Annarita Marrano, John McNamara, Sushma Naithani, Monica F Poelchau, Leonore Reiser and Margaret E. Staton (2024). *The AgBioData Consortium: FAIRification of Agricultural Data Through Standardization of Practices and Protocols Across Agricultural Databases.*. Plant & Animal Genome Conference/PAG31. San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Marcela K. Tello-Ruiz (2026). *The AgBioData Consortium: Ten Years of Coordinating FAIR Data Standards for Agricultural Genomics*. Plant Biology 2026. Ottawa, Canada. Status = OTHER; Acknowledgement of Federal Support = Yes

Annarita Marrano (2024). *The AgBioData Journey Towards Best Practices Of Data Sharing And Management In Agricultural Research And Education*. Plant Biology 2024. Hawaii, USA. Status = OTHER; Acknowledgement of Federal Support = Yes

Leonore Reiser (2022). *The AgBioData Research Coordination Network: Ensuring FAIR agricultural data through community-based standards*. AG2PI Workshop “Harnessing the Ag Genomics Data Torrent: A Community-driven Discussion on Best

Practices for Using and Reusing Genomics Data". Virtual. Status = OTHER; Acknowledgement of Federal Support = Yes

Annarita Marrano, Darwin Campbell, Jaqueline Campbell, Ethalinda Cannon, Laurel Cooper, Lisa Harper, Eva Huala, Sook Jung, Monica Poelchau, Leonore Reiser, Meg Staton, Marcela K.Tello-Ruiz (2022). *The AgBioData Research Coordination Network: Ensuring FAIR agricultural data through community-based standards*. The 2022 North American Plant Phenotyping Network Annual Conference. Virtual. Status = OTHER; Acknowledgement of Federal Support = Yes

Annarita Marrano, Darwin Campbell, Jaqueline Campbell, Ethalinda Cannon, Laurel Cooper, Lisa Harper, Eva Huala, Sunita Kumari, Sook Jung, Monica Poelchau, Leonore Reiser, Meg Staton, Marcela K Tello-Ruiz (2022). *The AgBioData Research Coordination Network: Ensuring FAIR agricultural data through community-based standards.*. The 2022 Plant Genome Online Conference. Virtual. Status = OTHER; Acknowledgement of Federal Support = Yes

Annarita Marrano, Darwin Campbell, Jaqueline Campbell, Ethalinda Cannon, Laurel Cooper, Lisa Harper, Eva Huala, Sunita Kumari, Sook Jung, Monica Poelchau, Leonore Reiser, Meg Staton, Marcela K Tello-Ruiz (2022). *The AgBioData Research Coordination Network: creating community-based standards for more findable, accessible, and reusable agricultural data*. The Bioinformatics Resources For Plant Biologists Workshop at the ASPB Plant Biology conference 2022. Virtual. Status = OTHER; Acknowledgement of Federal Support = Yes

Annarita Marrano, presenter (2023). *The AgBioData Research Coordination Network: creating community-based standards for more findable, accessible, and reusable agricultural data.*. American Society for Plant Biology. Savannah,GA, USA. Status = OTHER; Acknowledgement of Federal Support = Yes

John McNamara (2023). *The bigger picture of phenomics: whole-system support for systems research, models and databases*. American Society of Animal Science Meeting. Albuquerque, New Mexico. Status = OTHER; Acknowledgement of Federal Support = Yes

John McNamara, Marcela K. Tello-Ruiz and Dorrie Main (2026). *Update on Agbiodata Database Resource Sustainability*. International Plant and Animal Genome Conference (PAG33). San Diego, CA. Status = OTHER; Acknowledgement of Federal Support = Yes

Other Products

View all datasets currently available in the NSF Public Access Repository for this award.
The results in the NSF Public Access Repository will include a comprehensive listing of all datasets recorded to date that are associated with this award.

View all workshop reports currently available in the NSF Public Access Repository for this award.

The results in the NSF Public Access Repository will include a comprehensive listing of all workshop report publications recorded to date that are associated with this award.

Other Publications

Patent Applications

Technologies or Techniques

Thesis/Dissertations

Websites or Other Internet Sites
AgBioData Database Finder Tool
https://www.agbiodata.org/databasefinder

The AgBioDatabase Finder is an online tool developed by the AgBioData Consortium. It allows researchers to easily discover and deposit agricultural genomic, genetic, and breeding data across 25 member databases spanning various crop and livestock species.

AgBioData GitRepo
https://github.com/AgBioData

Git Hub Repository for AgBioData generated code and some working group products. It contains software including the Genome Assembly and Gene Model (GAAM) Identifier Tool. This command-line tool facilitates the creation and validation of genome assembly and gene model identifiers based on specific patterns.

AgBioData Recommended Repositories
https://www.agbiodata.org/db-finder-repositories

AgBioData's recommended primary repositories. For data types that are not directly submitted to AgBioData databases.

AgBioData YouTube Channel
https://www.youtube.com/agbiodataconsortium748/featured

AgBioData Consortium You Tube Channel. Recordings of our public webinars are posted here with the permission of the speaker.

AgBioData Zenodo Community
https://zenodo.org/communities/agb-nsf-rcn/records

The AgBioData consortium actively shares its open source curriculum materials, publications, and white papers on Zenodo. This is the long term archive for products.

AgBioData website
https://www.agbiodata.org/

The AgBioData website is the official site for the agricultural biological databases consortium. The website is the primary venue for sharing project objectives and deliverables. Key contents include published works, the agricultural FAIR educational curriculum, the AgBioDatabase Finder Tool to locate specific agricultural databases, monthly webinar listings and recordings.

Participants/Organizations

What individuals have worked on the project?

Name	Most Senior Project Role	Nearest Person Month Worked
Reiser, Leonore	PD/PI	2
Andorf, Carson	Co PD/PI	1
Main, Doreen	Co PD/PI	1
Molik, David	Co PD/PI	1
Poelchau, Monica	Co PD/PI	1
Clarke, Jennifer	Faculty	0
Elsik, Christine	Faculty	0
Imker, Heidi	Faculty	0
Jung, Jinha	Faculty	0
Ko, Dae Kwan	Faculty	0

Name	Most Senior Project Role	Nearest Person Month Worked
Koltes, James	Faculty	0
Lachowiec, Jennifer	Faculty	0
Larmande, Pierre	Faculty	0
Li, Qi	Faculty	0
McCarthy, Fiona	Faculty	0
McNamara, John	Faculty	0
Naithani, Sushma	Faculty	1
Whetzel, Trish	Faculty	0
Beier, Sebastian	Postdoctoral (scholar, fellow or other postdoctoral position)	0
Gupta, Parul	Postdoctoral (scholar, fellow or other postdoctoral position)	0
Haley, Olivia	Postdoctoral (scholar, fellow or other postdoctoral position)	0
Herminia Quezada Rodríguez, Elsa	Postdoctoral (scholar, fellow or other postdoctoral position)	0
Tibbs-Cortes, Laura	Postdoctoral (scholar, fellow or other postdoctoral position)	0
Jung, Sook	Staff Scientist (doctoral level)	0
Kumari, Sunita	Staff Scientist (doctoral level)	1
Tello-Ruiz, Marcela	Staff Scientist (doctoral level)	12
Callwood, Jodi	Graduate Student (research assistant)	0
Celebioglu, Burcu	Graduate Student (research assistant)	0
Kapoor, Muskan	Graduate Student (research assistant)	0
Kiat, Edwin	Graduate Student (research assistant)	0
Lin, Teng-Jui	Graduate Student (research assistant)	0

Name	Most Senior Project Role	Nearest Person Month Worked
Zhang, Junxiao	Graduate Student (research assistant)	0
Barrett, Gabriel	Other	0
Bassil, Nahla	Other	0
Becker, Doreen	Other	0
Berardini, Tanya	Other	0
Bushakra, Jill	Other	0
Cabugos, Leyla	Other	0
Campbell, Jacqueline	Other	0
Canon, Ethy	Other	0
Cezard, Timothee	Other	0
Chitneedi, Praveen	Other	0
Chougule, Kapeel	Other	0
Cole, Benjamin	Other	0
Cooper, Laurel	Other	0
Corleis, Björn	Other	0
Cottingham, Bob	Other	0
Daenzer, Jenna	Other	0
DASH, SUDHANSU	Other	0
Davenport, Kimberly	Other	0
Dyer, Sarah	Other	0
Giuffra, Elisabetta	Other	0
Goutte-Gattat, Damien	Other	0
Greenblum, Sharon	Other	0
Harb, Omar	Other	0

Name	Most Senior Project Role	Nearest Person Month Worked
Hu, Zhiliang	Other	0
Humann, Jodi	Other	0
Kapoor, Mehak	Other	0
Karaikal, Srikanth	Other	0
Lucic, Ana	Other	0
Meyers, Natalie	Other	0
Nelson, Rex	Other	0
Ng, Theros	Other	0
Olson, Andrew	Other	0
Pasha, Asher	Other	0
Poretsky, Elly	Other	0
Proia, Alyssa	Other	0
Raciti, Daniela	Other	0
Reyes Salinas, Fabian	Other	0
Richards, Stephen	Other	0
Selby, Peter	Other	0
Sen, Taner	Other	0
Shaw, Felix	Other	0
Sheehan, Moira	Other	0
Sibley, Gwen	Other	0
Tuggle, Chris	Other	0
Ware, Doreen	Other	0
Wei, Sharon	Other	0
Wright, Adam	Other	0

Name	Most Senior Project Role	Nearest Person Month Worked
Yook, Karen	Other	0

Full details of individuals who have worked on the project:

Leonore Reiser
Email: lreiser@arabidopsis.org
Most Senior Project Role: PD/PI
Nearest Person Month Worked: 2

Contribution to the Project: PD/PI, Steering Committee Member, Chair of Fair Scientific Literature Working Group.

Funding Support: Phoenix Bioinformatics and this grant (10%)

Change in active other support: No

International Collaboration: No
International Travel: No

Carson M Andorf
Email: carson.andorf@usda.gov
Most Senior Project Role: Co PD/PI
Nearest Person Month Worked: 1

Contribution to the Project: Co PI, member of steering committee

Funding Support: USDA

Change in active other support: No

International Collaboration: No
International Travel: No

Doreen Main
Email: dorrie@wsu.edu
Most Senior Project Role: Co PD/PI
Nearest Person Month Worked: 1

Contribution to the Project: Co PI, Chair of Sustainability Working Group, member of steering commitee,website management and hosting.

Funding Support: Washington State University

Change in active other support: No

International Collaboration: No
International Travel: No

David Molik
Email: molik@ksu.edu
Most Senior Project Role: Co PD/PI
Nearest Person Month Worked: 1

Contribution to the Project: Co PI, Lead organizer for Hybrid Conference, Member of Fair sceintific literature working group and member of steering committee.

Funding Support: Kansas State University

Change in active other support: Yes

International Collaboration: No
International Travel: No

Monica F Poelchau
Email: monica.poelchau@usda.gov
Most Senior Project Role: Co PD/PI
Nearest Person Month Worked: 1

Contribution to the Project: Co PI, Member of Steering Committee, Award review committee

Funding Support: USDA

Change in active other support: No

International Collaboration: No
International Travel: No

Jennifer Clarke
Email: jclarke3@unl.edu
Most Senior Project Role: Faculty
Nearest Person Month Worked: 0

Contribution to the Project: Co-chair of the Phenotypic Data Standardization and Management WG; Sustainability WG participant

Funding Support: University of Nebraska-Lincoln

International Collaboration: No
International Travel: No

Christine Elsik
Email: elsikc@missouri.edu
Most Senior Project Role: Faculty
Nearest Person Month Worked: 0

Contribution to the Project: AI-Readiness WG participant

Funding Support: University of Missouri

International Collaboration: No
International Travel: No

Heidi Imker
Email: imker@illinois.edu
Most Senior Project Role: Faculty
Nearest Person Month Worked: 0

Contribution to the Project: AgBioData Community Workshop participant (invited panelist), not supported by project

Funding Support: University of Illinois Urbana-Champaign

International Collaboration: No
International Travel: No

Jinha Jung
Email: jinha@purdue.edu
Most Senior Project Role: Faculty
Nearest Person Month Worked: 0

Contribution to the Project: Chair of the Phenotypic Data Standardization and Management WG, NLP4Biocuration WG and AI-Readiness WG participant

Funding Support: Purdue University

International Collaboration: No
International Travel: No

Dae Kwan Ko
Email: dkko@msu.edu
Most Senior Project Role: Faculty
Nearest Person Month Worked: 0

Contribution to the Project: AgBioData Community Workshop participant

Funding Support: Michigan State University

International Collaboration: No
International Travel: No

James Koltes
Email: jekoltes@iastate.edu
Most Senior Project Role: Faculty
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG and Sustainability WG participant, not supported by project

Funding Support: Iowa State University

International Collaboration: No
International Travel: No

Jennifer Lachowiec
Email: jennifer.lachowiec@montana.edu
Most Senior Project Role: Faculty
Nearest Person Month Worked: 0

Contribution to the Project: Phenotypic Data Standardization and Management WG participant

Funding Support: Department of Plant Sciences and Plant Pathology, Montana State University, Bozeman, MT

International Collaboration: No
International Travel: No

Pierre Larmande
Email: pierre.larmande@ird.fr
Most Senior Project Role: Faculty
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG participant

Funding Support: IRD (Institute of Research for Sustainable Development)

International Collaboration: Yes, France
International Travel: No

Qi Li
Email: qli@iastate.edu
Most Senior Project Role: Faculty
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG participant

Funding Support: Iowa State University

International Collaboration: No
International Travel: No

Fiona McCarthy
Email: fionamcc@arizona.edu
Most Senior Project Role: Faculty
Nearest Person Month Worked: 0

Contribution to the Project: Sustainability WG participant

Funding Support: University of Arizona

International Collaboration: No
International Travel: No

John McNamara
Email: mcnamara@wsu.edu
Most Senior Project Role: Faculty
Nearest Person Month Worked: 0

Contribution to the Project: Steering Commitee until May 2026, Co-Chair of the Sustainability WG, Phenotypic Data Standardization & Management WG and AI-Readiness WG participant, Ambassador (received funding to attend PAG33)

Funding Support: Washington State University

International Collaboration: No
International Travel: No

Sushma Naithani
Email: Sushma.Naithani@oregonstate.edu
Most Senior Project Role: Faculty
Nearest Person Month Worked: 1

Contribution to the Project: Steering Committee, Co-chair of the Education & Training WG, NLP4Biocuration WG and Sustainability WG participant, Ambassador (received funding to attend the 2026 ISB Conference)

Funding Support: Oregon State University

International Collaboration: No
International Travel: No

Trish Whetzel
Email: trish@tislab.org
Most Senior Project Role: Faculty
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG participant

Funding Support: University of North Carolina at Chapel Hill

International Collaboration: No
International Travel: No

Sebastian Beier
Email: s.beier@fz-juelich.de
Most Senior Project Role: Postdoctoral (scholar, fellow or other postdoctoral position)
Nearest Person Month Worked: 0

Contribution to the Project: Standards for Genetic Variation WG participant, not supported by project

Funding Support: ELIXIR Plant Science Community, Research Centre Jülich

International Collaboration: Yes, Germany
International Travel: No

Parul Gupta
Email: parul.gupta@oregonstate.edu
Most Senior Project Role: Postdoctoral (scholar, fellow or other postdoctoral position)
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG participant,

Funding Support: Oregon State University

International Collaboration: No
International Travel: No

Olivia Haley
Email: Olivia.Haley@usda.gov
Most Senior Project Role: Postdoctoral (scholar, fellow or other postdoctoral position)
Nearest Person Month Worked: 0

Contribution to the Project: New member of the Steering Committee, Co-chair of the AIR WG, NLP4Biocuration WG, Education & Training WG and AgBioData Community Workshop participant, received funding through the Champion Program

Funding Support: USDA-ARS

International Collaboration: No
International Travel: No

Elsa Herminia Quezada Rodríguez
Email: qrelsa@gmail.com
Most Senior Project Role: Postdoctoral (scholar, fellow or other postdoctoral position)
Nearest Person Month Worked: 0

Contribution to the Project: Phenotypic Data Standardization and Management WG and AgBioData Community Workshop (virtual) participant

Funding Support: Universidad Autónoma Metropolitana

International Collaboration: Yes, Mexico
International Travel: No

Laura Tibbs-Cortes
Email: laura.tibbs-cortes@usda.gov
Most Senior Project Role: Postdoctoral (scholar, fellow or other postdoctoral position)
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG participant, received funding through the Champion Program

Funding Support: USDA-ARS

International Collaboration: No
International Travel: No

Sook Jung
Email: sook_jung@wsu.edu
Most Senior Project Role: Staff Scientist (doctoral level)
Nearest Person Month Worked: 0

Contribution to the Project: FAIR Scientific Literature WG and NLP4Biocuration WG participant, not supported by project, former co-PI

Funding Support: Washington State University

International Collaboration: No
International Travel: No

Sunita Kumari
Email: kumari@cshl.edu
Most Senior Project Role: Staff Scientist (doctoral level)
Nearest Person Month Worked: 1

Contribution to the Project: Steering Committee, Co-chair of the Single Cell RNA-Seq Biocuration WG, AI-Readiness WG and AgBioData Community Workshop participant, Ambassador (received funding to attend the 2026 ASPB Conference)

Funding Support: CSHL/Gramene

International Collaboration: No
International Travel: No

Marcela Karey Tello-Ruiz
Email: marcela@phxbio.org
Most Senior Project Role: Staff Scientist (doctoral level)
Nearest Person Month Worked: 12

Contribution to the Project: Project Role: Project or program coordinator. Coordination of the working group, workshop preparation, webinar organization, support to the steering committee, planning and preparation of educational material, recruiting activities, outreach and conference presentations. Standards for Genetic Variation WG, NLP4Biocuration WG, AI-Readiness WG, Sustainabity WG and Education & Training WG participant

Funding Support: Funding support is 100% from this grant.

International Collaboration: No
International Travel: No

Jodi Callwood
Email: jodicall@iastate.edu
Most Senior Project Role: Graduate Student (research assistant)
Nearest Person Month Worked: 0

Contribution to the Project: Phenotypic Data Standards Management WG, NLP4Biocuration WG and AgBioData Community Workshop participant

Funding Support: Iowa State University

International Collaboration: No
International Travel: No

Burcu Celebioglu
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Most Senior Project Role: Graduate Student (research assistant)
Nearest Person Month Worked: 0

Contribution to the Project: Phenotypic Data Standards Management WG participant

Funding Support: Oregon State University

International Collaboration: No
International Travel: No

Muskan Kapoor
Email: muskan@iastate.edu
Most Senior Project Role: Graduate Student (research assistant)
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant, Ambassador (received funding to attend PAG33)

Funding Support: Iowa State University

International Collaboration: No
International Travel: No

Edwin Ong Jun Kiat
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Most Senior Project Role: Graduate Student (research assistant)
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG participant

Funding Support: Queen's University of Belfast

International Collaboration: Yes, Ireland
International Travel: No

Teng-Jui (Owen) Lin
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Most Senior Project Role: Graduate Student (research assistant)
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: UC Berkeley

International Collaboration: No
International Travel: No

Junxiao Zhang
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Most Senior Project Role: Graduate Student (research assistant)
Nearest Person Month Worked: 0

Contribution to the Project: Phenotypic Data Standardization and Management WG participant

Funding Support: University of Nebraska-Lincoln

International Collaboration: No
International Travel: No

Gabriel Barrett
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG and AI-Readiness WG and AgBioData Community Workshop participant

Funding Support: University of Connecticut

International Collaboration: No
International Travel: No

Nahla Bassil
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: Standards for Genetic Variation WG participant

Funding Support: USDA-ARS

International Collaboration: No
International Travel: No

Doreen Becker
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: Research Institute for Farm Animal Biology (FBN)

International Collaboration: Yes, Germany
International Travel: No

Tanya Berardini
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: PI of member database, Sustainability WG and AgBioData Community Workshop participant, Chair of the NLP4Biocuration WG, received funding through the Champion and Ambassador (for PAG33) programs

Funding Support: Phoenix Bioinformatics

International Collaboration: No
International Travel: No

Jill Bushakra
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: Standards for Genetic Variation WG participant

Funding Support: USDA-ARS

International Collaboration: No
International Travel: No

Leyla Cabugos
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: FAIR Scientific Literature WG participant, received funding through the Champion Program

Funding Support: California Polytechnic State University, San Luis Obispo

International Collaboration: No
International Travel: No

Jacqueline Campbell
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG and FAIR Scientific Literature WG participant

Funding Support: USDA-ARS

International Collaboration: No
International Travel: No

Ethy Canon
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: AI-Readiness WG participant

Funding Support: USDA-ARS

International Collaboration: No
International Travel: No

Timothee Cezard
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: Standards for Genetic Variation WG Co-Chair, not supported by project

Funding Support: EMBL/EBI

International Collaboration: Yes, United Kingdom
International Travel: No

Praveen Krishna Chitneedi
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: Research Institute for Farm Animal Biology (FBN)

International Collaboration: Yes, Germany
International Travel: No

Kapeel Chougule
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG and AI-Readiness WG participant, Ambassador (received funding to attend the 2026 AGBT-Ag Conference)

Funding Support: CSHL/Gramene

International Collaboration: No
International Travel: No

Benjamin Cole
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: LBL

International Collaboration: No
International Travel: No

Laurel Cooper
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG participant

Funding Support: Oregon State University

International Collaboration: No
International Travel: No

Björn Corleis
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: Friedrich Loeffler Institut

International Collaboration: Yes, Germany
International Travel: No

Bob Cottingham
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG participant

Funding Support: Oak Ridge National Laboratory

International Collaboration: No
International Travel: No

Jenna Daenzer
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: FAIR Scientific Literature WG participant

Funding Support: Genetics Society of America

International Collaboration: No
International Travel: No

SUDHANSU DASH
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG participant

Funding Support: NCGR (NATIONAL CENTER FOR GENOME RESOURCES)

International Collaboration: No
International Travel: No

Kimberly M. Davenport
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: Department of Animal Sciences, Washington State University

International Collaboration: No
International Travel: No

Sarah Dyer
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: Steering Commitee Member until May 2026, Co-chair of the AI-Readiness WG from Jan-May 2026, not supported by project

Funding Support: EMBL/EBI

International Collaboration: Yes, United Kingdom
International Travel: No

Elisabetta Giuffra
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: INRAE

International Collaboration: Yes, France
International Travel: No

Damien Goutte-Gattat
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: University of Cambridge / FlyBase

International Collaboration: Yes, United Kingdom
International Travel: No

Sharon Greenblum
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: LBL

International Collaboration: No
International Travel: No

Omar Harb
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: AgBioData Community Workshop participant (invited panelist)

Funding Support: University of Pennsylvania

International Collaboration: No
International Travel: No

Zhiliang Hu
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG participant

Funding Support: Iowa State University

International Collaboration: No
International Travel: No

Jodi Humann
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: Standards for Genetic Variation WG participant

Funding Support: Washington State University

International Collaboration: No
International Travel: No

Mehak Kapoor
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: Bioinformatics and Computational Biology Program, Department of Animal Science, Iowa State University

International Collaboration: No
International Travel: No

Srikanth Kumar Karaikal
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG participant

Funding Support: Cornell University

International Collaboration: No
International Travel: No

Ana Lucic
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: Phenotypic Data Standardization and Management WG participant

Funding Support: University of Illinois Urbana-Champaign

International Collaboration: No
International Travel: No

Natalie Meyers
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: AgBioData Community Workshop participant (invited panelist)

Funding Support: San Diego Supercomputer Center

International Collaboration: No
International Travel: No

Rex Nelson
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG and AI-Readiness WG participant

Funding Support: SoyBase, USDA-ARS

International Collaboration: No
International Travel: No

Theros T Ng
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: College of Veterinary Medicine, Western University of Health Sciences

International Collaboration: No
International Travel: No

Andrew Olson
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG and AI-Readiness WG participant

Funding Support: CSHL

International Collaboration: No
International Travel: No

Asher Pasha
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: AgBioData Community Workshop (virtual) participant, not supported by project

Funding Support: University of Toronto

International Collaboration: Yes, Canada
International Travel: No

Elly Poretsky
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG participant

Funding Support: USDA-ARS

International Collaboration: No
International Travel: No

Alyssa Proia
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: NLP4Biocuration WG participant

Funding Support: Phoenix Bioinformatics

International Collaboration: No
International Travel: No

Daniela Raciti
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: FAIR Scientific Literature WG participant

Funding Support: microPublication Biology Journal/California Institute of Technology

International Collaboration: No
International Travel: No

Fabian Andres Reyes Salinas
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: Department of Horticultural Sciences, University of Florida, Gainesville FI

International Collaboration: No
International Travel: No

Stephen Richards
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: AgBioData Community Workshop participant (invited panelist)

Funding Support: University of California, Davis

International Collaboration: No
International Travel: No

Peter Selby
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: Phenotypic Data Standardization and Management WG participant, not supported by project

Funding Support: Cornell University

International Collaboration: No
International Travel: No

Taner Sen
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: Steering Committee Member, NLP4Biocuration WG participant

Funding Support: USDA-ARS

International Collaboration: No
International Travel: No

Felix Shaw
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: Earlham Institute

International Collaboration: Yes, United Kingdom
International Travel: No

Moira Sheehan
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: Standards for Genetic Variation WG participant

Funding Support: University of Florida

International Collaboration: No
International Travel: No

Gwen Sibley
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: AgBioData Community Workshop participant

Funding Support: Kansas State University

International Collaboration: No
International Travel: No

Chris Tuggle
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: scRNA Biocuration WG participant

Funding Support: Iowa State University

International Collaboration: No
International Travel: No

Doreen Ware
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: Standards for Genetic Variation WG, NLP4Biocuration, AI-Readiness and scRNA-Seq Biocuration WG participant

Funding Support: USDA-ARS

International Collaboration: No
International Travel: No

Sharon Wei
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: Standards for Genetic Variation WG, AI-Readiness WG, and AgBioData Community Workshop participant

Funding Support: CSHL/SorghumBase

International Collaboration: No
International Travel: No

Adam Wright
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Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: FAIR Scientific Literature WG participant, Co-Chair of the NLP4Biocuration WG

Funding Support: Ontario Institute for Cancer Research

International Collaboration: Yes, Canada
International Travel: No

Karen Yook
Email: karen.yook@micropublication.org
Most Senior Project Role: Other
Nearest Person Month Worked: 0

Contribution to the Project: FAIR Scientific Literature WG participant

Funding Support: microPublication Biology Journal/California Institute of Technology

International Collaboration: No
International Travel: No

What other organizations have been involved as partners?

Name	Type of Partner Organization	Location
Cold Spring Harbor Laboratory	Academic Institution	Cold Spring Harbor, NY
EMBL-EBI	Other Organizations (foreign or domestic)	Hinxton, Cambridgeshire, UK
Oregon State University	Academic Institution	Corvallis, OR
Washington State University	Academic Institution	Pullman, WA

Full details of organizations that have been involved as partners:

Cold Spring Harbor Laboratory

Organization Type: Academic Institution
Organization Location: Cold Spring Harbor, NY

Partner's Contribution to the Project:
Collaborative Research

More Detail on Partner and Contribution: Sunita Kumari is a steering committee member.

EMBL-EBI

Organization Type: Other Organizations (foreign or domestic)
Organization Location: Hinxton, Cambridgeshire, UK

Partner's Contribution to the Project:
Collaborative Research

More Detail on Partner and Contribution: Peter Harrison is a steering committee member.

Oregon State University

Organization Type: Academic Institution
Organization Location: Corvallis, OR

Partner's Contribution to the Project:
Collaborative Research

More Detail on Partner and Contribution: Sushma Naithani is a steering committee member.

Washington State University

Organization Type: Academic Institution
Organization Location: Pullman, WA

Partner's Contribution to the Project:
Collaborative Research

More Detail on Partner and Contribution: Sook Jung is CoPi, Dorrie Main and John McNamara are steering committee members.

Were other collaborators or contacts involved? If so, please provide details.

As stated in Aim 2, AgBioData continues to expand its network by engaging with projects and organizations that share common interests in agricultural data, standards, and sustainability. We establish connections through conferences, invited webinar presentations, community workshops, and working group activities. This year, invited webinar speakers included representatives from Gene Ontology, VEuPathDB, and NCBI's Gene Expression Omnibus (GEO) and representatives from VEuPathDB participated in the sustainability panel at our community workshop.

We also maintain a close partnership with the USDA-NIFA-funded Agricultural Genomes to Phenomes Initiative (AG2PI), with reciprocal representation on our external advisory boards. Through our external advisory board and individual members, we maintain connections with the RDA Alliance, USDA funded NRSP8, FAIR Sharing, and Alliance for Genome Resources.

In addition, we collaborate with the Functional Annotation of Animal Genomes (FAANG) and Plant Cell Atlas (PCA) communities to promote FAIR data principles and community standards for agricultural single cell data.

Impacts

What is the impact on the development of the principal discipline(s) of the project?

The year 5 activities advanced AgBioData's FAIR-data work through deployable standards, shared tools, and AI-readiness practices for agricultural genomics, genetics, breeding, and informatics. The Standards for Genetic Variation Working Group published a peer-reviewed standard and FAIR user guide, and nearly 220 million reference SNP identifiers were assigned to agricultural variants. Five active working groups now address AI readiness, AI-driven literature curation, single-cell RNA-seq data standardization, and database sustainability. The NLP4Biocuration group created a shared manuscript-curation platform that brings multiple USDA-ARS projects together through one interface and common infrastructure, reducing duplicated development and supporting more consistent curation. The AI-Readiness group is also exploring a shared connection layer that could let AI assistants search multiple specialized organism databases through one standardized interface, avoiding a custom connection for every database. Together, these advances make public agricultural data more useful for cross-database analysis and AI, supporting the national objective to build high-quality AI-ready scientific datasets and USDA's Farmers First emphasis on research that delivers practical value to American agriculture.

What is the impact on other disciplines?

AgBioData's year 5 products extend beyond agricultural biology because its FAIR practices, curricula, and shared tools can be reused wherever complex scientific data must be organized and connected. The work supports library and information science instruction, research-data services, and computer and data science teams that prepare and combine information for databases, knowledge graphs, and AI tools that retrieve evidence from trusted sources. The AgBioDatabase Finder Tool and repository list help authors, publishers, librarians, and curators curate data in stable, appropriate repositories, while the FAIR Scientific Literature Working Group helps publishers understand the value of data curation and community databases. The shared manuscript-curation platform shows how one set of tools can serve multiple USDA-ARS projects, and is easily scalable to other scientific disciplines. The proposed common AI connection would similarly let researchers and AI tools search many specialized organism databases through one interface instead of building a separate link to each one. By standardizing data, these efforts support America's AI Action Plan goal of creating high-quality, AI-ready scientific datasets. They also make it easier to connect information about organisms, experiments, environments, hosts, pests, pathogens, and phenotypes, strengthening public-health preparedness, environmental monitoring, agricultural surveillance, and risk assessment. Open curricula and professional training extend these practices to students and researchers in other fields, building a workforce able to apply them wherever complex data are generated. This cross-disciplinary capacity supports USDA's Farm Security National Security initiative and the priority to protect American agriculture from invasive pests and diseases.

What is the impact on the development of human resources?

The project expanded and strengthened the agricultural data workforce during Year 5. AgBioData recruited 39 new members, and working-group participation engaged 73 people. Three Champion Awards recognized leaders of major deliverables, seven Ambassador Awards supported representation at four conferences, and nine community awards recognized service and leadership. Participants gained experience in standards, AI-ready data management, biocuration, scientific communication, and project leadership. The shared manuscript-curation platform also gives participants from multiple USDA-ARS projects practical experience with common workflows and cross-project collaboration, while exploration of a common AI connection builds skills spanning biology, databases, and artificial intelligence. Webinars, hybrid participation, conference sessions, and open materials broadened access to this professional development. These activities create a workforce able to manage trustworthy scientific data and apply AI responsibly, supporting the Administration's emphasis on developing an AI-ready workforce and building technical capacity for U.S. agricultural research and rural industries.

What was the impact on teaching and educational experiences?

The seven-lesson AgBioData Curriculum for FAIR Ag Science has now been downloaded from Zenodo more than 2,000 times. The new FAIR User Guide for adoption of standard SNP identifiers gives instructors and trainees a current, practical case study in making agricultural variant data interoperable and reusable. AgBioData also maintained an open monthly webinar archive; the YouTube channel grew to 211 subscribers, with an average of 31 views per webinar. The 2026 hybrid community workshop attracted 76 registrants or interested participants, and more than 96% of evaluation respondents were satisfied or very satisfied; 81% reported a better understanding of AgBioData working groups. Together, these resources give undergraduate, graduate, and professional learners hands-on exposure to data stewardship, standards, biocuration, and AI readiness, strengthening the scientific and technical workforce needed for American leadership in agricultural innovation and AI-enabled science.

What is the impact on physical resources that form infrastructure?

The RCN did not purchase or construct major physical infrastructure during Year 5; its principal infrastructure impacts were digital and operational. The shared manuscript-curation platform brings multiple USDA-ARS projects onto a common interface and computing infrastructure, reducing the need for each project to maintain a separate system. AgBioData standards, metadata guidance, and repository recommendations also help laboratories and database teams use existing computing and storage more efficiently by reducing later cleaning, reformatting, and repeated repository-ingest work. The common AI connection being explored could further reduce duplication by giving multiple databases the same basic pathway for interacting with AI assistants. Together, these shared approaches support efficient, cost-conscious research infrastructure and increase the long-term value of publicly funded agricultural data.

What is the impact on institutional resources that form infrastructure?

This RCN’s year 5 outcomes provided institutions with evidence and practical tools for sustaining agricultural community databases. The project completed impact-assessment summaries covering 77% of AgBioData's crop and livestock databases and held individual discussions with leaders from 20 of 26 primary databases to identify common sustainability needs. A five-

database pilot evaluated measures such as use, citations, economic value, and return on public investment, while the sustainability workshop at the Plant and Animal Genome Conference brought users, managers, and developers together to identify shared solutions. Strategic briefs for government, funding, institutional, and industry audiences are translating these findings into actionable cases for support. Together, these advances help institutions prioritize shared services, reduce duplication, and preserve high-value data resources that support Farmers First and rural economic productivity.

What is the impact on information resources that form infrastructure?

This RCN's year 5 activities strengthened agricultural information infrastructure. The published genetic-variation standard, FAIR user guide, and assignment of nearly 220 million reference SNP identifiers make variant records easier to connect and reuse across studies and databases. The AgBioDatabase Finder and recommended-repository list improve data submission and discovery.. The NLP4Biocuration group established a shared platform and maize prototype that brings multiple USDA-ARS projects together for machine-assisted literature curation. The AI-Readiness group is developing a database assessment framework and exploring a shared connection layer that could allow AI assistants to search multiple organism databases consistently. Impact assessments covering 77% of member crop and livestock databases also provide evidence for sustaining these resources. Collectively, these advances create more connected, durable, and machine-actionable agricultural data assets and support America's AI Action Plan goal of building high-quality AI-ready scientific datasets.

What is the impact on technology transfer?

The project accelerated transfer by converting community consensus into products that can be adopted without proprietary software or licensing barriers: a peer-reviewed reference-SNP standard, a concise FAIR user guide, the AgBioDatabase Finder, a repository-selection resource, an open curriculum, and shared AI-assisted curation prototypes. AgBioData disseminated these products through workshops, webinars, its redesigned website, seven Ambassadors at four conferences, and presentations or exhibits at five national and international meetings. The sustainability team also prepared targeted communications for government, institutional, and industry stakeholders. These activities reduce the time and expertise organizations need to standardize, integrate, and reuse agricultural data. For example, AgBioData principles have guided new USDA-ARS efforts to build FAIR-compliant pest and pathogen databases, which support national goals to safeguard plant and animal health. By providing faster access to interoperable evidence, it can shorten the path from public research to markers, traits, surveillance tools, and decision support, thereby supporting USDA priorities to increase producer profitability, protect agriculture, and expand the value of U.S. agricultural products.

What is the impact on society beyond science and technology?

AgBioData delivers public value by making federally and institutionally funded agricultural research easier to find, verify, combine, and reuse. The Year 5 standards, repository guidance, website improvements, and sustainability assessments protect the value of public investments in databases that steward genetics, genomics, and breeding data across crop and livestock research. AI-ready, interoperable data can reduce the time required to identify useful traits, compare crop germplasm and livestock breeding populations, investigate pests and diseases, and generate evidence relevant to food quality and nutrition. Open curricula, webinars, and leadership programs also prepare scientists and data professionals for an AI-enabled economy, addresses the bioliteracy gap identified by the National Security Commission on Emerging Biotechnology, and supports the "Empower American Workers in the Age of AI" in America's AI Action Plan through practical training in data and AI literacy. These outcomes support Farmers First and Rural Revival by strengthening the research foundation for productive, competitive agricultural industries; Farm Security is National Security by improving the information base for plant and animal health protection; and the USDA priority of improving human health through food quality. The result is a more capable workforce and a more reliable agricultural knowledge system serving producers, researchers, and the public.

Although this work is grounded in agriculture, its effects extend across the food system and the broader economy. FAIR agricultural data and advanced AI capabilities support a safe and reliable food supply, earlier detection of plant and animal health threats, more resilient supply chains, rural economic vitality, and continued U.S. competitiveness in research and technology. By preparing researchers and other professionals to manage, interpret, and responsibly apply agricultural data, the project strengthens the knowledge infrastructure that underpins public health, biosecurity, environmental stewardship, and national security.

What percentage of the award's budget was spent in a foreign country?

Nothing to report.

Changes/Problems

Changes in approach and reason for change

We originally planned to organize the hybrid conference through Washington State University at a third-party site, as we did in 2022, but changed the subcontract and venue to Kansas State University. Dr. David Molik is a faculty member there and joined the Steering Committee in April 2025. This provided an opportunity for us to host the meeting at a centrally located site on the campus of a land grant university with the potential to engage with the local community, which was something we had wanted to do from the start. We requested and received permission for all of these changes.

Actual or Anticipated problems or delays and actions or plans to resolve them
Nothing to report.

Changes that have a significant impact on expenditures

While no changes have significantly impacted expenditures, we spent less than budgeted for the hybrid conference. Several factors contributed to this. The first was changing the subcontract from Washington State to Kansas State which eliminated some of the organizational costs. Second, Dr. Molik was able to source additional sponsorship for meals thereby reducing grant expenses. Third, a lower than projected number of in-person participants.

Significant changes in use or care of human subjects
Nothing to report.

Significant changes in use or care of vertebrate animals
Nothing to report.

Significant changes in use or care of biohazards
Nothing to report.

Change in primary performance site location
Nothing to report.

Special Requirements

Responses to any special reporting requirements specified in the award terms and conditions, as well as any award specific reporting requirements.
Nothing to report.